

Prompting Fundamentals on Real Tasks v2

Track: General AI Fluency

Assignment Code: FL-02

Task Chosen: Building my FlyRank ML Internship project (Content Refresh Opportunity Scoring)

Task

For this assignment, I chose my real internship project from my FL-01 Workflow Audit. The task is to frame my Machine Learning project correctly before building any model. The goal is to create a refresh priority score that helps content teams decide which pages should be reviewed first using search and engagement data.

Version 0 – Naive Prompt

Prompt

Help me frame my ML project.

Output

Machine Learning projects usually begin by defining the problem, selecting features, choosing an algorithm, and evaluating performance. Consider whether your task is classification, regression, or clustering.

Reflection

This is the type of prompt I would have written before starting the internship. The response explained general Machine Learning concepts but did not relate to my actual project. It lacked context, structure, and practical guidance.

Version 1 – Role Assignment

Technique Used

Role Assignment

Prompt

You are an experienced Machine Learning mentor. Help me frame my ML project.

Output

As your ML mentor, I recommend defining the business problem before selecting any algorithm. Start by identifying the decision your model will support, then determine the appropriate learning task.

Reflection

Adding a role changed the style of the response. Instead of simply defining Machine Learning concepts, the AI started giving practical advice. However, it still didn't know anything about my internship project, so the suggestions remained fairly generic.

Version 2 – Context and Motivation

Technique Used

Context and Motivation

Prompt

You are an experienced Machine Learning mentor. I am working on the FlyRank ML Internship. My project is Content Refresh Opportunity Scoring. I want to rank website pages based on search and engagement signals so a content team knows which pages should be reviewed first. Help me frame the project correctly.

Output

Your project is a decision-support problem rather than a prediction for its own sake. The objective is to produce a ranked list of pages that deserve review. Before considering Machine Learning, define the target, the unit of analysis, and create a simple baseline that can later be compared against ML methods.

Reflection

Providing project context produced the biggest improvement so far. The AI stopped giving generic ML explanations and instead focused on my actual assignment. The advice became much more relevant to the work I was doing.

Version 3 – Few-Shot Examples

Technique Used

Few-Shot Examples

Prompt

You are an experienced Machine Learning mentor.

Example:

Project: Email Spam Detection

Task Type: Classification

Target: Spam or Not Spam

Decision: Filter unwanted emails

Now help me frame my project.

Project: Content Refresh Opportunity Scoring

Goal: Rank website pages that should be reviewed first.

Output

Task Type: Scoring / Ranking

Target: Refresh Priority Score

Decision Supported: Which pages should be reviewed first.

Reason: The project combines multiple performance signals into a ranked priority list instead of predicting a simple yes or no outcome.

Reflection

The example showed the AI exactly how I wanted the answer presented. The response became shorter, more consistent, and easier to use in my notebook. Instead of guessing the format, the AI followed the pattern provided.

Version 4 – Output Structure

Technique Used

Output Structure

Prompt

You are an experienced Machine Learning mentor. I am working on the FlyRank ML Internship. My project is Content Refresh Opportunity Scoring, where I want to rank pages that should be reviewed first based on search and engagement signals.

Present your answer using the following headings:

- ML Task Type
- Target
- Unit of Analysis
- Success Metric
- Why ML Instead of Rules
- Next Steps

Output

ML Task Type

Scoring / Ranking

Target

Refresh Priority Score

Unit of Analysis

One row represents one content page.

Success Metric

An understandable and explainable ranking that helps content teams prioritize pages for review.

Why ML Instead of Rules

Multiple signals interact with each other, making fixed thresholds less reliable than a learned scoring approach.

Next Steps

Build a simple rule-based baseline first, then compare it with ML models.

Reflection

Adding a required output structure made the response much easier to use. Instead of reading a long paragraph, I received information already organized according to my notebook sections. This reduced the amount of editing I needed to do before using the response.

Version 5 – Step Decomposition

Technique Used

Step Decomposition

Prompt

You are an experienced Machine Learning mentor helping with the FlyRank ML Internship.

My project is Content Refresh Opportunity Scoring.

Help me one step at a time.

Step 1: Identify the ML task.

Step 2: Define the target.

Step 3: Define the unit of analysis.

Step 4: Explain why ML is appropriate.

Step 5: Suggest the next implementation steps.

Do not skip steps or combine them.

Output

Step 1

This is a scoring problem because each page receives a priority score.

Step 2

The target is a Refresh Priority Score representing how urgently a page should be reviewed.

Step 3

The unit of analysis is one content page represented by one row in the dataset.

Step 4

Several performance signals interact, making the decision difficult to capture with one simple rule.

Step 5

Create a rule-based baseline, evaluate it, then compare its performance with Machine Learning methods.

Reflection

Breaking the task into smaller steps produced the clearest response of all six prompts. Every part of the assignment was answered individually, making it easier to understand and verify. This approach also reduced the chance that the AI would skip important information.

Cross-Model Comparison

After completing the final prompt, I tested it in both Claude and ChatGPT.

Claude

Claude produced a structured response that closely followed the requested steps. It stayed focused on the internship context and avoided making assumptions about work I had not completed. The explanations were detailed and emphasized careful reasoning before choosing a model.

ChatGPT

ChatGPT also followed the requested structure but explained the concepts in simpler language. The response was easier for me to understand as a student and included practical examples. In a few places it suggested additional ideas, but these were clearly presented as recommendations rather than facts.

Overall Comparison

Both models produced useful responses, but they had different strengths. Claude was stronger at maintaining strict structure and following instructions exactly. ChatGPT was stronger at simplifying complex ideas and making them easier to understand. For this assignment, I found

ChatGPT slightly more helpful while learning the concepts, while Claude was better when I needed formal documentation.

Final Reusable Prompt Template

You are an experienced **[ROLE]** helping with **[PROJECT NAME]**.

Project Context

[Describe the project, available data, current progress, and overall objective.]

Task

Help me complete the following task:

[TASK]

Instructions

Work through the problem step by step.

Do not make assumptions about work I have not completed.

If information is missing, clearly state what assumptions you are making.

Use simple language suitable for a Computer Science student.

Output Format

1. Problem
2. Task Type
3. Target
4. Unit of Analysis
5. Success Metric
6. Recommendations
7. Next Steps